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1. A. Firstly, I created an IAM role in AWS to establish administrative permissions through a custom trust policy.

C. From there, I initiated the role creation process and selected "Custom trust policy" as the trusted entity type, allowing for more granular control over access permissions.

Permissions policies (1/1045) [info](#)

Choose one or more policies to attach to your new role.

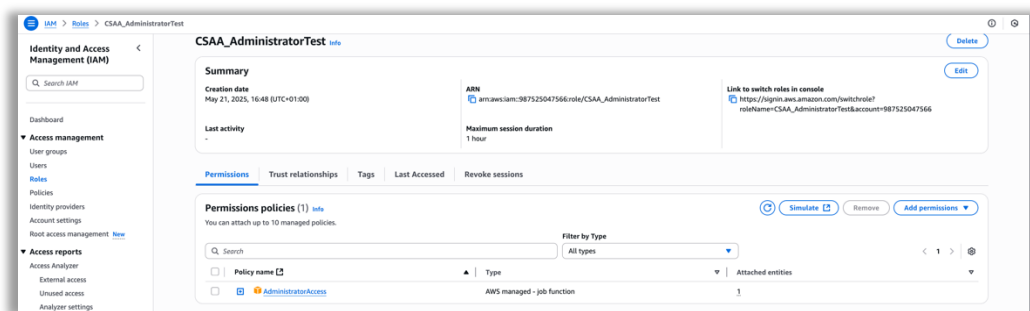
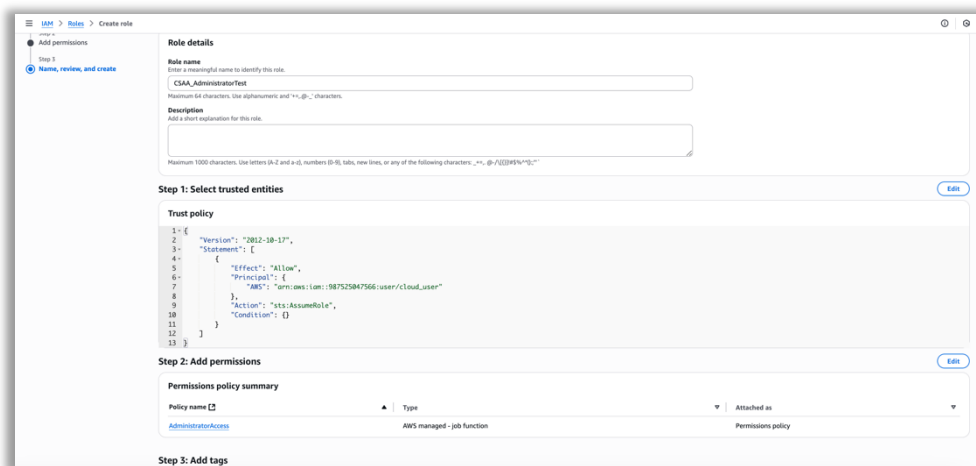
Search

Policy name	Type	Description
☒ AdministratorAccess	AWS managed - job function	Provides full access to AWS services and resources.
☐ AdministratorAccess-Ampify	AWS managed	Grants administrator permissions while explicitly allowing direct access to resources needed by Ampify applications.
☐ AdministratorAccess-AWSCloudRe...	AWS managed	Grants account administrative permissions. Explicitly allows developers and administrators to grant direct access to resources they need to manage A...
☐ AWSCloudAdminPolicy	AWS managed	Provides ReadOnly permissions required by the Amazon AI Operations Assistant to do analysis on customer AWS resources during investigations.
☐ AWSCloudConsoleAdminPolicy	AWS managed	Grants full access to Amazon AI Operations service and its required permissions via AWS console. It also includes permissions to use identity-aware c...
☐ AWSCloudOperationsAccess	AWS managed	Grants access to the Amazon AI Operations APIs for creating, updating, and deleting investigations, investigation events, and investigation resource...
☐ AWSCloudReadOnlyAccess	AWS managed	Grants ReadOnly permissions to the Amazon AI Operations service and its required resources.
☐ AlexaforBusinessDeviceSetup	AWS managed	Provide device setup access to AlexaforBusiness services
☐ AlexaforBusinessFullAccess	AWS managed	Grants full access to AlexaforBusiness resources and access to related AWS Services
☐ AlexaforBusinessGatewayExecution	AWS managed	Provide gateway execution access to AlexaforBusiness services
☐ AlexaforBusinessFeatureDelegate...	AWS managed	Provide access to Lefuse AWS devices
☐ AlexaforBusinessPushOutgoingSto...	AWS managed	Provide access to Pushy AWS devices
☐ AlexaforBusinessReadOnlyAccess	AWS managed	Provide read only access to AlexaforBusiness services
☐ allow_all	Customer managed	-
☐ AmazonAPIGatewayAdministrator	AWS managed	Provides full access to create/delete/update APIs in Amazon API Gateways via the AWS Management Console.
☐ AmazonAPIGatewayInvokeFullAccess	AWS managed	Provides full access to invoke APIs in Amazon API Gateway.
☐ AmazonAPIGatewayPushToCloudL...	AWS managed	Allows API Gateway to push logs to user's account.
☐ AmazonAmazonTimestreamFullAccess	AWS managed	Provides full access to Amazon Timestream and access to AWS services supported as flow source or destination (S3 and Redshift). Also provides access ...
☐ AmazonAmazonTimestreamReadOnlyAccess	AWS managed	Provides read only access to Amazon Timestream flows

2. A. Once the IAM role was successfully created, I proceeded to assume it in order to verify that the permissions were correctly configured. If the role was not already open from the previous step, I located the “CSAA_AdministratorTest” role from the IAM roles list within the AWS Console and opened it.

B. From the role's Summary section, I copied the Switch Role URL provided by AWS. I then opened a new browser tab and navigated to that URL, which automatically pre-filled the required fields such as the Account ID and the Role ARN.

C. For easier identification, I optionally assigned a display name and selected a colour theme. Finally, I selected Switch Role, which confirmed that the role was being assumed successfully within the same AWS account. This step ensured that I could work under the newly created administrative permissions for further configuration and testing.



aws

English

Switch Role

Switching roles enables you to manage resources across Amazon Web Services accounts using a single user. When you switch roles, you temporarily take on the permissions assigned to the new role. When you exit the role, you give up those permissions and get your original permissions back. [Learn more](#)

Account ID

The 12-digit account number or the alias of the account in which the role exists.

987525047566

IAM role name

The name of the role that you want to assume which can be found at the end of the role's ARN. For example, provide the `TestRole` role name from the following role ARN: `arn:aws:iam::123456789012:role/TestRole`.

CSAA_AdministratorTest

Display name - optional

This name will appear in the console navigation bar when active. Choose a name to help identify the permission set assigned to the role.

CSAA_AdministratorTest @ 987525047566

Display color - optional

The selected color displays in the console navigation when this role is active

☐ None

Cancel

Switch Role

aws

Search

[Option+S]

Europe (Stockholm)

CSAA_AdministratorTest @ 987525047566

Console Home

Reset to default layout

+ Add widgets

Recently visited

IAM

CloudFormation

Infrastructure Composer

View all services

Applications (0)

Region: Europe (Stockholm)

Select Region

eu-north-1 (Current Region)

Find applications

Name

Description

Region

Originati...

1

Access denied to servicecatalog:ListApplications

Diagnose with Amazon Q

Go to myApplications

Welcome to AWS

AWS Health

Cost and usage

IAM > Roles

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

Users

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Policies

Identity providers

Account settings

Root access management

Access reports

Access Analyzer

External access

Unused access

Analyzer settings

Credential report

Organization activity

Service control policies

Resource control policies

IAM Identity Center

AWS Organizations

Roles (31)

1 match

Role name

Trusted entities

Last activity

CSAA_AdministratorTest

Account: 987525047566

4 minutes ago

Roles Anywhere

Authenticate your non AWS workloads and securely provide access to AWS services.

Access AWS from your non AWS workloads

Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.

X.509 Standard

Use your own existing PKI infrastructure or use AWS Certificate Manager Private Certificate Authority to authenticate identities.

Temporary credentials

Use temporary credentials with ease and benefit from the enhanced security they provide.

3. A. After verifying that the IAM role was functioning as expected, I moved on to codifying it using AWS CloudFormation to enable consistent, repeatable deployments in the future.

B. I began by opening a new tab and navigating to the CloudFormation service within the AWS Management Console. Under the Stacks section, I selected Create stack and chose -with new resources (standard) as the creation method.

C. To prepare the template, I opted to use Application Composer, selecting Build from Application Composer under the "Prepare template" prerequisites. Once the Application Composer environment launched, I clicked Create in Application Composer to initiate the process.

D. In the upper section of the interface, I accessed the Template view and selected YAML as my preferred template language (though JSON was also available). I then pasted the relevant CloudFormation code for the IAM role into the editor and clicked Create Template.

E. Before doing so, I made sure to copy the IAM Trust Policy from the lab's GitHub repository and replaced %REPLACE_WITH_ACCOUNT_ID% with my actual AWS Account ID. This trust policy ensures that only the IAM identity associated with the cloud_user ARN is allowed to assume the role, following the principle of least privilege.

F. Next, I confirmed the transfer bucket name prompt by selecting Confirm and continue to CloudFormation, which redirected me back to the CloudFormation workflow.

G. I proceeded through the configuration by selecting Next, entered AdministratorAccessRole as the stack name, and continued by clicking Next again. While optional, tag information could be added at this point to support better resource organization.

H. On the final review screen, I acknowledged that CloudFormation might create IAM resources with custom names by checking the relevant box. I then selected Submit to initiate deployment of the stack.

I. The template was deployed successfully. I monitored the stack's progress and refreshed the view until the status showed CREATE_COMPLETE, confirming the role was now available for future use.

CloudFormation

Stacks

AdministratorAccessRole

Stacks

Stack details

Drifts

StackSets

Exports

Infrastructure Composer

laC generator

Hooks overview

Hooks

Registry

Public extensions

Activated extensions

Publisher

Stacks (1)

Filter by stack name

Filter status

Active

View nested

Stacks

AdministratorAccessRole

2025-05-21 19:16:34 UTC+0100

CREATE_COMPLETE

AdministratorAccessRole

Delete

Update stack

Stack actions

Create stack

Stack info

Events

Resources

Outputs

Parameters

Templ

Table view

Timeline view

Events (5)

View root cause

Search events

Timestamp	Logical ID	Status	Detailed
2025-05-21 19:16:56 UTC+0100	AdministratorAccessRole	CREATE_COMPLETE	-
2025-05-21 19:16:55 UTC+0100	CSAAAdministratorRole	CREATE_COMPLETE	-
2025-05-21 19:16:37 UTC+0100	CSAAAdministratorRole	CREATE_IN_PROGRES 5	-

IAM

Roles

CSAA_Administrator

Identity and Access Management (IAM)

Search IAM

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Root access management

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External access

Unused access

Analyzer settings

Credential report

Organization activity

Service control policies

Resource control policies

IAM Identity Center

CSAA_Administrator

IAM role with Administrator access

Summary

Creation date

May 21, 2025, 19:16 (UTC+01:00)

ARN

arn:aws:iam::680829786361:role/CSAA_Administrator

Last activity

-

Maximum session duration

1 hour

Link to switch roles in console

https://signin.aws.amazon.com/switchrole?roleName=CSAA_Administrator&account=680829786361

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Trusted entities

Entities that can assume this role under specified conditions.

```
1- {
2-   "Version": "2012-10-17",
3-   "Statement": [
4-     {
5-       "Effect": "Allow",
6-       "Principal": {
7-         "AWS": "arn:aws:iam::680829786361:user/cloud_user"
8-       },
9-       "Action": "sts:AssumeRole"
10-     }
11-   ]
12- }
```